

Similar improvements in inhibitory control following low-volume high-intensity interval exercise and moderate-intensity continuous exercise

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ABSTRACT

Objectives: We previously demonstrated that a traditional high-volume (HV) high-intensity interval exercise (HIIE) was more effective in improving post-exercise inhibitory control (IC) than moderate-intensity continuous exercise (MCE). Nevertheless, because HV-HIIE is performed with a higher volume and longer duration, it may result in decreased adherence to this exercise for some individuals. In this study, we compared the effect of a practical low-volume (LV) HIIE to that of MCE on post-exercise IC improvements.

Design: Twenty healthy males performed both LV-HIIE and MCE on a cycle ergometer in a crossover design.

Method: LV-HIIE was consisted of ten 1-min bouts at 90% of the peak oxygen consumption ($\text{VO}_{2\text{ peak}}$) with 1-min active recovery at 30% of $\text{VO}_{2\text{ peak}}$, which had a total duration of 20 min. MCE was performed for 40 min at 60% of $\text{VO}_{2\text{ peak}}$. To evaluate IC, the Stroop test was administered before exercise, immediately after exercise, and every 10 min during the 30-min post-exercise recovery period.

Results: IC significantly improved immediately after LV-HIIE and MCE compared with that before each exercise (both P s < 0.05). The improved IC remained significant until the 20-min post-exercise recovery period for both protocols (all P s < 0.05). The degrees of post-exercise IC improvements throughout the 30-min post-exercise recovery period did not differ significantly between protocols.

Conclusions: These findings demonstrated that despite lower exercise volume and shorter exercise duration, LV-HIIE could improve post-exercise IC similar to MCE.

1. Introduction

Cognitive dysfunction, as seen in mild cognitive impairment and cognitive function disorder, is a prominent factor that indicates poor prognosis in older individuals and patients with chronic diseases (DeCarli, 2003). Hence, prevention of cognitive dysfunction is as urgent issue in public health worldwide. Cognitive executive function involves three characteristics: shifting of mental sets, monitoring and updating of working memory representations, and inhibition of prepotent responses (Miyake et al., 2000). The latter is often referred to as Inhibitory control (IC) and defined as the suppression of behavior in response to either an internal or external stimulus (Ozonoff & Strayer, 1997), which is necessary to prevent the implementation of an unrequired action (Coxon, Stinear, & Byblow, 2007). Long-term intervention of aerobic

exercise improved cognitive executive function, including IC, in various populations (Hillman, Erickson, & Kramer, 2008). Additionally, we previously reported that an acute bout of aerobic exercise could improve post-exercise IC (Hashimoto et al., 2018; Tanaka et al., 2018; Tsukamoto et al., 2016a,b, 2017, 2018) and the degree of post-exercise IC improvements is associated with an increase in aerobic exercise variables such as exercise intensity, exercise duration, and exercise volume (i.e., the product of exercise intensity and duration; Tsukamoto et al., 2016a, 2017). Furthermore, we determined that the degree of post-exercise IC improvements was greater following moderate-intensity exercise (i.e., 60% $\text{VO}_{2\text{ peak}}$ for 20 min) than following volume-matched low-intensity exercise (i.e., 30% $\text{VO}_{2\text{ peak}}$ for 40 min; Tsukamoto et al., 2017). This finding suggests that when performing aerobic continuous exercise, exercise intensity may play a more important role in improving

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post-exercise IC rather than exercise duration and exercise volume.

Long-term intervention of a traditional high-volume (HV) high-intensity interval exercise (HIIE) is known to be more effective in improving many physical functions, including exercise performance and cardiovascular and metabolic health, than that of moderate-intensity continuous exercise (MCE) in various populations (Tjønnå et al., 2008; Wisløff et al., 2007). We previously demonstrated that HV-HIIE, which consisted of four bouts of 4-min high-intensity exercise with 3-min active recovery, was more effective in sustaining post-exercise IC improvements than volume-matched MCE (Tsukamoto et al., 2016a). Thus, HV-HIIE is more capable of improving not only physical functions but also cognitive function than MCE. However, because this HIIE type is performed with high exercise volume and long exercise duration, it may result in reduced adherence to this exercise among some individuals (Trost, Owen, Bauman, Sallis, & Brown, 2002).

Alternatively, low-volume (LV) HIIE is performed with a lower volume and shorter duration than traditional HV-HIIE and MCE. LV-HIIE involves repeated “all-out” maximal-intensity cycling effort (Gibala, Little, MacDonald, & Hawley, 2012), which is also referred to as sprint-interval exercise (Hardcastle, Ray, Beale, & Hagger, 2014; Stork, Kwan, Gibala, Martin & Ginis, 2015). This exhaustive type of LV-HIIE induces feelings of severe fatigue and decreases the affective response and exercise motivation (Stork, Kwan, Gibala, & Martin Ginis, 2015), which may lead to a reduction in adherence among some individuals (Hardcastle et al., 2014). By contrast, practical LV-HIIE is performed without “all-out” maximal-intensity efforts but consists of a shorter active recovery period time than HV-HIIE (e.g., ten bouts of 1-min high-intensity exercise with 1-min active recovery) (Little, Safdar, Wilkin, Tarnopolsky, & Gibala, 2010). Previous studies determined that a long-term intervention of this LV-HIIE type is more effective in improving some physical functions than that of MCE in various populations (Hood, Little, Tarnopolsky, Myslik, & Gibala, 2011; Little et al., 2010, 2011). Furthermore, previous studies determined that exercise enjoyment, which is a major parameter related to exercise adherence (Trost et al., 2002), may be higher following an acute bout of LV-HIIE than following an acute bout of MCE (Jung, Bourne, & Little, 2014; Thum, Parsons, Whittle, & Astorino, 2017). However, no such effect is expected from HV-HIIE (Decker & Ekkekakis, 2017; Sagelv et al., 2019), potentially because it has higher volume and longer duration than LV-HIIE (Martinez, Kilpatrick, Salomon, Jung, & Little, 2015). Therefore, the LV-HIIE may be a useful protocol for improving adherence to HIIE, while potentially conferring similar training adaptations to HV-HIIE (Hood et al., 2011; Little et al., 2010, 2011). Nevertheless, to the best of our knowledge, no study has compared the effect of LV-HIIE and MCE on post-exercise IC improvements.

Previous studies reported that an increase in blood lactate level was greater following LV-HIIE than following MCE (Parker et al., 2017; Poon, Sheridan, Chung, & Wong, 2018) and may be similar to increases following HV-HIIE (Olney et al., 2018). Our previous studies demonstrated that increased levels of circulating lactate induced by aerobic exercise are related to the degree of post-exercise IC improvements (Hashimoto et al., 2018; Tsukamoto et al., 2016b), potentially due to an increase in cerebral lactate metabolism (Hashimoto et al., 2018). Thus, we hypothesized that the degree of exercise-induced IC improvements would be greater following LV-HIIE than following MCE. To test this hypothesis, we compared the effect of LV-HIIE and MCE on post-exercise IC improvements.

2. Methods

2.1. Subjects and ethics

Twenty healthy young males (age: 22.2 ± 0.4 years, body height: 173.0 ± 1.0 cm, body mass: 66.8 ± 2.1 kg, $\text{VO}_{2\text{ peak}}$: 44.3 ± 1.3 ml/min/kg) participated in this study. The subjects were recreationally active and participated in physical exercise (e.g., aerobic exercise) for 2–4 h

per week, but were not involved in any specific physical training programs and did not belong to any sports teams within the previous 3 years. The subjects provided written informed consent upon having the experimental procedures and potential risks described to them. As the Stroop test was performed using the subject's right hand, all subjects were right-hand dominant, which was ascertained by asking each subject which hand they preferred to use for writing. Furthermore, they were free of any known neurological, cardiovascular, and pulmonary disorders, as well as free from color-blindness and abnormal vision. The subjects were instructed to avoid strenuous physical activity in the 24 h prior to each experimental session. Each subject also abstained from food, caffeine and alcohol for 12 h prior to each experiment, and were not taking any medications that may affect cognitive performance. All procedures were approved by the Ethics Committee of Ritsumeikan University (BKC-IRB-2014-028).

2.2. Experimental design

All subjects completed a familiarization visit where they practiced a minimum of 10 trials for each task of the Stroop test until they achieved consistent scores. During this visit, the subjects also performed a ramp-incremental test on a cycle ergometer to determine their $\text{VO}_{2\text{ peak}}$, which was used to calculate the workload for LV-HIIE and MCE protocols.

On the day of the experiment, the subjects practiced the three tasks of the Stroop tests before exercise to minimize a learning effect. The subjects practiced each Stroop test a minimum of five times. Following this, the subjects rested for 5 min before undergoing measurements of cardiovascular parameters and blood metabolites. Once this was completed (i.e., 10-min after the Stroop test practice session), the subjects performed the pre-exercise Stroop test. Subsequently, the subjects completed either the LV-HIIE or MCE protocol. The Stroop test was performed again immediately after completion of the exercise session, and then was repeated three times at 10-min intervals during the 30-min post-exercise recovery period in order to evaluate the prolonged effects of post-exercise improvements of IC.

During exercise, cardiovascular parameters, rate of perceived exertion (RPE), and felt arousal scale (FAS) were measured intermittently on five occasions for each protocol as follows; LV-HIIE was measured every 2 sets (during active recovery) and MCE was measured every 8 min. The data of the five occasions of cardiovascular parameters and RPE during the exercise session were averaged and used for analyses. Additionally, cardiovascular parameters were measured before exercise and during the 30-min post-exercise recovery period (i.e., three times at 10-min intervals). Blood samples were collected after each cardiovascular measurement. FAS and visual analogue scale (VAS) were recorded immediately after the Stroop test throughout the experimental session.

To account for the learning effect on improvements in IC, in a preliminary study, we examined the effect of staying in a sitting rest for 45 min on IC in ten healthy young males (age: 22.4 ± 0.3 years, body height: 173.1 ± 1.9 cm, body mass: 67.3 ± 2.3 kg). According to the experimental design of the main study, the Stroop test was performed before the condition, immediately after the condition, and every 10 min during 30-min after the condition. The results showed that response times and accuracies of the three Stroop tasks and the reverse-Stroop interference score IC did not significantly differ among the five time points (data not shown). Thus, we considered that the results of exercise-induced IC improvements obtained in this study were not attributable to a learning effect.

2.3. Ramp-incremental test

During the familiarization visit, all subjects performed a ramp-incremental exercise test on a cycle ergometer to determine $\text{VO}_{2\text{ peak}}$. The subjects performed 3 min of baseline cycling at 30 W, after which the workload was increased at a rate of 30 W/min until the limit of tolerance. The subjects were asked to maintain a cadence of 60 rpm

throughout the test. During the incremental test, breath-by-breath pulmonary gas exchange data were collected using a gas analyzer (AE-310S; Minato Medical Science, Osaka, Japan) and averaged every 10 s. $\text{VO}_{2\text{ peak}}$ was determined by the highest 30-s mean value attained prior to exhaustion. The criteria for exhaustion have been described previously (Tanaka et al., 2018; Tsukamoto et al., 2016a,b, 2017, 2018).

2.4. Exercise protocols

LV-HIIE and MCE protocols used in this study are presented in Figure 1. All subjects completed LV-HIIE or MCE conditions on a cycle ergometer in a randomized and counterbalanced order. The two experimental days were separated by a week interval for all subjects, with approximately same experiment start time (± 1 h).

To ensure that the Stroop test was conducted over the same time interval for both LV-HIIE and MCE sessions, subjects completed a 20 min rest period in a seated position without doing any other tasks (e.g., paperwork, office work) that may affect cognitive function before the LV-HIIE to account for its shorter exercise duration. The exercise protocol in each condition was performed following a warm-up at 50 W for 5 min. After the rest period and warm-up, LV-HIIE protocol incorporated ten 1-min bouts at 90% of peak oxygen uptake ($\text{VO}_{2\text{ peak}}$) with 1-min active recovery at 30% of $\text{VO}_{2\text{ peak}}$, which totaled 20 min in duration. MCE protocol was performed at 60% of $\text{VO}_{2\text{ peak}}$ and lasted 40 min in duration (Tsukamoto et al., 2016a, 2017). The subjects were instructed to maintain a cadence of 60 rpm throughout exercise session for each protocol, which was carefully monitored by an examiner.

2.5. Cardiovascular parameters

Heart rate (HR) and mean arterial pressure (MAP) were collected to determine cardiovascular responses. HR was measured continuously via telemetry (RS400; Polar Electro Japan, Tokyo, Japan). Systolic blood pressure (SBP) and diastolic blood pressure (DBP) were measured using a mercury manometer (FC-110ST; Focal, Chiba, Japan). MAP was calculated as $[(\text{SBP} - \text{DBP})/3 + \text{DBP}]$.

2.6. Blood metabolites

Fingertip blood samples were collected to determine blood metabolite responses. Blood glucose levels were measured using a glucose analyzer (Glutest Neo α ; Sanwa Kagaku Kenkyusho, Nagoya, Japan).

Blood lactate levels were measured using a lactate analyzer (Lactate Pro 2; Arkray, Kyoto, Japan).

2.7. Perceived exertion

RPE was measured to assess the perceived exertion (i.e., physiological response) during exercise. RPE is a 15-point scale ranging from 6 (no exertion) to 20 (maximal exertion) (Borg, 1982).

2.8. FAS-measured arousal level

FAS-measured arousal was measured to assess the activation–arousal (i.e., psychological response) during exercise and for the Stroop test. FAS is a 6-point, single-item scale ranging from 1 (low-arousal) to 6 (high arousal) (Svebak & Murgatroyd, 1985).

2.9. VAS-measured psychological conditions

VAS was measured to assess the psychological conditions for the Stroop test. This consists of questions that assess mental fatigue, the ability to concentrate, and motivation. Each VAS was labeled from 0 mm (i.e., not at all) to 100 mm (i.e., extremely). The subjects drew lines to indicate their response.

2.10. IC

The Stroop test was administered to determine IC, which is a well-known paradigm for investigating domains of cognitive performance that depend on IC (Stroop, 1935). The stimulus words of the Stroop test were consisted of four color names (“RED”, “YELLOW”, “GREEN” and “BLUE”), and they were presented on a 98-inch display. All words were written in Japanese for Japanese subjects. The subjects were required to press the color-labeled keypad that corresponded to the text meaning of the stimulus word. The Stroop test consisted of two color text tasks and one control black text task. The congruent task, which is a facilitated task, displayed the color names presented in the same-colored text. The neutral task, which is a control task, displayed the color names presented in black text. The incongruent task, which is an interference task, displayed the color names presented in a different-colored text. One trial of each task consisted of 24 stimulus words, which were presented in a random order. The Stroop test for each time point consisted of three trials for each task in a random and counterbalanced order. The reaction

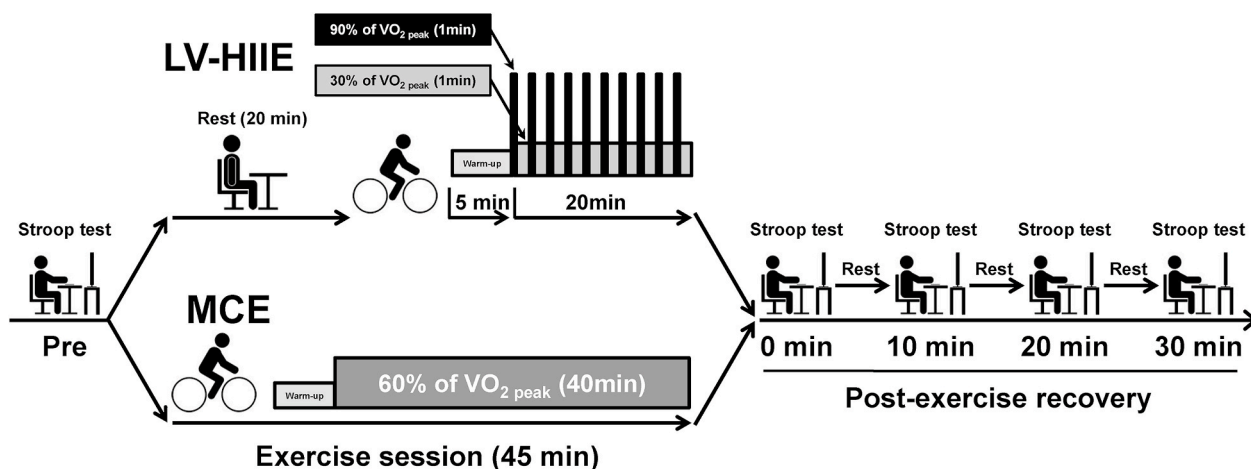


Fig. 1. Experimental procedures of low-volume high intensity exercise (LV-HIIE) and moderate-intensity continuous exercise (MCE) sessions. The LV-HIIE session employed a rest period of 20 min in a sitting position following the pre-exercise Stroop test. After the rest period, the LV-HIIE protocol was performed with ten 1-min bouts at 90% of peak oxygen uptake ($\text{VO}_{2\text{ peak}}$) with 1-min active recovery at 30% of $\text{VO}_{2\text{ peak}}$, which was 20 min in duration. The MCE protocol was performed at 60% of $\text{VO}_{2\text{ peak}}$ for 40 min. The Stroop test was administered before exercise, immediately after exercise, and every 10 min during the 30-min post-exercise recovery period.

time and response accuracy for each trial were collected and the average of the three trials for each of the three Stroop tests was calculated. IC assessed using the reverse-Stroop interference score, which is the difference between the reaction times for the neutral and incongruent tasks. The reverse-Stroop interference score was calculated as [(reaction time of incongruent task – reaction time of neutral task)/reaction time of neutral task × 100] (Ikeda, Hirata, Okuzumi, & Kokubun, 2010).

2.11. Statistical analysis

Prior to the present study, we calculated the required sample size based on an α -level of 0.05 and a β -level of 0.2 (80% power) to detect statistical significance of a two-way (2 conditions × 5 times) repeated-measures analysis of variance (ANOVA) of the primary outcome variable (i.e., the reverse-Stroop interference score) using a G power software (version 3.1.9.2, Germany). The effect size used a partial eta squared (η_p^2) value (0.30) of a significant interaction effect of the reverse-Stroop interference score between LV-HIIE and MCE obtained in our previous study (Tsukamoto et al., 2016a). As the result, a required minimum number of subjects was 15; therefore, we considered that a subject number recruited in this study was sufficient for ensuring statistical power and sensitivity of this study.

The data are expressed as the mean ± SEM. Comparison of exercise volume, cardiovascular parameters, and RPE between LV-HIIE and MCE was performed using a paired Student's *t*-test. Changes in reaction times and response accuracies of the Stroop test throughout experimental session were analyzed by using a three-way (task type × condition × time) repeated-measures ANOVA. Changes in other variables throughout experimental session were analyzed using a two-way (condition × time) repeated-measures ANOVA. If the sphericity assumption was not met, Greenhouse-Geisser corrections were used. The η_p^2 values were determined as a measure of the effect size for main effects of conditions and time and interaction effect. Specific differences were identified via a Bonferroni *post-hoc* test. Significance was established at $P < 0.05$. All statistical analyses were conducted using IBM SPSS software (Ver. 19.0, IBM Corp, NY, USA).

The Cohen's *d* effect size using the pooled standard deviation was calculated to determine the magnitude of a difference in measured variables between time points. The Cohen's *d* effect size were interpreted as small (0.20–0.49), medium (0.50–0.79), and large (>0.80; Cohen, 1992).

3. Results

3.1. Measured variables during exercise session

Exercise volume was nearly two-fold lower in LV-HIIE protocol than in MCE (176 ± 4 vs. 347 ± 9 kJ, $P < 0.001$). With regard to cardiovascular responses, mean HR during exercise was significantly higher for LV-HIIE than for MCE (166 ± 4 vs. 158 ± 4 bpm; $P < 0.001$, $d = 0.52$). In contrast, mean MAP during exercise did not differ between LV-HIIE and MCE (99 ± 3 vs. 97 ± 2 mmHg). With regard to physiopsychological and psychological responses, mean RPE during exercise was significantly higher for LV-HIIE than for MCE (15.5 ± 0.4 vs. 14.9 ± 0.3, $P < 0.01$, $d = 0.46$). In contrast, mean arousal during exercise did not differ between LV-HIIE and MCE (4.0 ± 0.2 vs. 4.0 ± 0.2).

3.2. Changes in physiological variables throughout experimental session

Changes in cardiovascular and blood metabolite responses throughout LV-HIIE and MCE sessions are shown in Table 1. With regard to cardiovascular responses, HR analysis revealed significant main effects for condition ($F_{(1,19)} = 9.69$, $P = 0.006$, $\eta_p^2 = 0.34$) and time ($F_{(1.81,34.40)} = 647.29$, $P < 0.001$, $\eta_p^2 = 0.97$), and a significant interaction effect ($F_{(4,76)} = 8.43$, $P < 0.001$, $\eta_p^2 = 0.31$). HR significantly increased immediately after both LV-HIIE and MCE compared with that before

each exercise (both P s < 0.001, $d = 7.72$ and 7.20, respectively), and the increased HR remained significant throughout the 30-min post-exercise recovery period for both protocols. HR values immediately after exercise and at the 20-min post-exercise recovery period were significantly higher for LV-HIIE than for MCE (both P s < 0.05, $d = 0.58$ and 0.41, respectively). MAP analysis revealed a significant main effect for time ($F_{(1.64,31.24)} = 6.08$, $P = 0.009$, $\eta_p^2 = 0.24$). Although there were no significant differences for MAP among all time points throughout LV-HIIE session, a trend against significance was observed for MAP between before and 30 min after MCE ($P = 0.054$, $d = 0.54$). With regard to blood metabolite responses, blood glucose analysis revealed a significant main effect for condition ($F_{(1,19)} = 11.97$, $P = 0.003$, $\eta_p^2 = 0.39$) and a significant interaction effect ($F_{(2.15,40.92)} = 5.83$, $P = 0.005$, $\eta_p^2 = 0.24$). A trend against significant main effect was observed for time in blood glucose analysis ($F_{(1.78,33.76)} = 2.85$, $P = 0.077$, $\eta_p^2 = 0.13$). Blood glucose immediately after MCE was significantly lower than that at 10 min after exercise ($P = 0.006$, $d = 0.44$). Blood glucose levels at 20 and 30 min after MCE were significantly lower than that before the exercise (both P s < 0.05, $d = 0.68$ and 0.71, respectively). A trend against significance was observed for blood glucose levels before and immediately after MCE ($P = 0.051$, $d = 0.87$). Blood lactate analysis revealed significant main effects for condition ($F_{(1,19)} = 38.0$, $P < 0.001$, $\eta_p^2 = 0.67$) and time ($F_{(1.30,24.61)} = 74.24$, $P < 0.001$, $\eta_p^2 = 0.80$), and a significant interaction effect ($F_{(1.42,26.91)} = 48.51$, $P < 0.001$, $\eta_p^2 = 0.72$). Blood lactate significantly increased immediately after both LV-HIIE and MCE compared with that before each exercise (both P s < 0.001, $d = 2.91$ and 2.09, respectively), and the increased blood lactate remained significant until the 30-min and 20-min post-exercise recovery periods for LV-HIIE and MCE, respectively (all P s < 0.05, $d = 1.09$ to 1.60). Blood lactate levels throughout the 30-min post-exercise recovery period were significantly higher for LV-HIIE than for MCE (all P s < 0.001, $d = 1.16$ to 1.91).

3.3. Changes in IC throughout experimental session

Changes in reaction times and response accuracies on three Stroop tests throughout LV-HIIE and MCE sessions are summarized in Table 2. Reaction time analysis for the Stroop test revealed significant main effects for task type ($F_{(1.38,26.14)} = 159.20$, $P < 0.001$, $\eta_p^2 = 0.89$) and time ($F_{(4,76)} = 23.70$, $P < 0.001$, $\eta_p^2 = 0.56$) and a significant interaction effect between task type and time ($F_{(4.27,81.05)} = 2.56$, $P = 0.041$, $\eta_p^2 = 0.12$). The main effect for task type indicated that reaction time of congruent task was significantly shorter than that of both neutral and incongruent tasks (both P s < 0.001) and reaction time of neutral task was significantly shorter than of that incongruent task ($P < 0.001$). With regard to time main effect-dependent analyses for each task, congruent reaction time significantly shortened immediately after both LV-HIIE and MCE compared with that before each exercise (both P s < 0.01, $d = 0.52$ and 0.37, respectively), and the shortened congruent reaction time remained significant until the 10-min post-exercise recovery periods for MCE ($P = 0.002$, $d = 0.26$). Neutral reaction time significantly shortened immediately after both LV-HIIE and MCE compared with that before each exercise (both P s < 0.05, $d = 0.30$ and 0.27, respectively). Incongruent reaction time significantly shortened immediately after both LV-HIIE and MCE compared with that before each exercise (both P s < 0.001, $d = 0.54$ and 0.47, respectively), and the shortened incongruent reaction time remained significant until the 20-min and 30-min post-exercise recovery periods for LV-HIIE and MCE, respectively (all P s < 0.05, $d = 0.17$ to 0.37). Response accuracy analysis for the Stroop test revealed a significant main effect for task type ($F_{(2,38)} = 3.74$, $P = 0.033$, $\eta_p^2 = 0.16$). The main effect for task type indicated that a trend against significance was observed for response accuracies between neutral and incongruent tasks ($P = 0.056$).

Changes in IC throughout LV-HIIE and MCE sessions are presented in Figure 2. The reverse-Stroop interference score analysis revealed a significant main effect for time ($F_{(4,76)} = 8.55$, $P < 0.001$, $\eta_p^2 = 0.31$). The

Table 1

Changes in cardiovascular and blood metabolite responses throughout experimental sessions in low-volume high-intensity interval exercise (LV-HIIE) and moderate-intensity continuous exercise (MCE).

	Time points					P values		
	Pre-EX	Post-EX 0	Post-EX 10	Post-EX 20	Post-EX 30	Condition	Time	Interaction
Heart rate (bpm)								
LV-HIIE	68.3 ± 1.6	174.3 ± 4.0 ^a	97.9 ± 3.3 ^{ab}	90.4 ± 3.1 ^{abc}	85.1 ± 2.7 ^{abcd}	0.006	<0.001	<0.001
MCE	68.4 ± 1.4	163.9 ± 3.9 ^{*a}	94.8 ± 3.1 ^{ab}	85.1 ± 2.8 ^{*abc}	80.9 ± 2.4 ^{abcd}			
MAP (mmHg)								
LV-HIIE	91.0 ± 1.6	96.7 ± 3.3	91.5 ± 1.3	90.6 ± 1.4	90.3 ± 1.2	0.463	0.009	0.642
MCE	91.5 ± 1.3	95.5 ± 2.4	90.6 ± 1.4	89.9 ± 1.1	88.4 ± 1.1			
Blood glucose (mg/dl)								
LV-HIIE	96.1 ± 1.9	97.1 ± 2.5	102.6 ± 3.5	98.2 ± 2.5	94.8 ± 2.2	0.003	0.077	0.005
MCE	96.6 ± 1.7	87.9 ± 2.6 [*]	93.1 ± 2.6 ^{*b}	91.4 ± 1.7 ^{*a}	90.6 ± 2.0 ^{*a}			
Blood lactate (mM)								
LV-HIIE	1.1 ± 0.1	9.8 ± 1.0 ^a	6.2 ± 0.9 ^{ab}	4.2 ± 0.7 ^{abc}	3.0 ± 0.4 ^{abcd}	<0.001	<0.001	<0.001
MCE	1.1 ± 0.1	3.6 ± 0.4 ^{*a}	2.1 ± 0.2 ^{*ab}	1.7 ± 0.2 ^{*abc}	1.4 ± 0.1 ^{*bcd}			

Values are presented as Mean ± SEM. MAP; mean arterial pressure, Pre-EX; before exercise, Post-EX 0; immediately after exercise, Post-EX 10; 10-min at post-exercise recovery period, Post-EX 20; 20-min at post-exercise recovery period, Post-EX 30; 30-min at post-exercise recovery period. Bold P values indicate a significant main effect of time and a significant interaction effect ($P < 0.05$ for all). * Significant difference ($P < 0.05$) between LV-HIIE and MCE. ^a Significant difference ($P < 0.05$) from Pre-EX. ^b Significant difference ($P < 0.05$) from Post-EX 0. ^c Significant difference ($P < 0.05$) from Post-EX 10. ^d Significant difference ($P < 0.05$) from Post-EX 20.

Table 2

Summary of reaction times and response accuracies of three Stroop tasks throughout LV-HIIE and MCE sessions.

	Reaction time (msec)		Response accuracy (%)	
	LV-HIIE	MCE	LV-HIIE	MCE
Congruent task				
Pre-EX	8675 ± 297	8712 ± 376	97.5 ± 0.7	97.0 ± 0.7
Post-EX 0	7965 ± 310 ^a	8127 ± 329 ^a	97.2 ± 0.6	96.9 ± 0.7
Post-EX 10	8275 ± 296	8266 ± 382 ^a	97.1 ± 0.8	97.0 ± 0.6
Post-EX 20	8342 ± 386	8335 ± 372	97.0 ± 0.6	96.8 ± 0.7
Post-EX 30	8376 ± 347	8594 ± 373	98.2 ± 0.4	97.5 ± 0.5
Neutral task				
Pre-EX	8878 ± 387	8934 ± 342	98.1 ± 0.5	97.6 ± 0.5
Post-EX 0	8413 ± 305 ^a	8530 ± 331 ^a	98.0 ± 0.5	97.6 ± 0.5
Post-EX 10	8625 ± 261	8589 ± 353	97.5 ± 0.5	97.9 ± 0.6
Post-EX 20	8657 ± 351	8757 ± 380	96.8 ± 0.8	97.7 ± 0.5
Post-EX 30	8710 ± 326	8896 ± 385	97.5 ± 0.7	97.9 ± 0.6
Incongruent task				
Pre-EX	9806 ± 373	9830 ± 388	96.5 ± 0.7	96.9 ± 0.6
Post-EX 0	8920 ± 358 ^a	9024 ± 376 ^a	96.8 ± 0.8	97.3 ± 0.6
Post-EX 10	9168 ± 300 ^a	9128 ± 394 ^a	97.6 ± 0.6	98.1 ± 0.4
Post-EX 20	9214 ± 341 ^a	9362 ± 409 ^a	97.2 ± 0.6	96.7 ± 0.8
Post-EX 30	9394 ± 351 ^b	9526 ± 390 ^{abc}	97.3 ± 0.5	97.3 ± 0.7

Values are presented as means ± SEM. ^a Significant difference ($P < 0.05$) from Pre-EX. ^b Significant difference ($P < 0.05$) from Post-EX 0. ^c Significant difference ($P < 0.05$) from Post-EX 10.

reverse-Stroop interference score significantly decreased immediately after both LV-HIIE and MCE compared with that before each exercise (both $P_s < 0.05$, $d = 1.14$ and 1.25 , respectively), and the decreased reverse-Stroop interference score remained significant until the 20-min post-exercise recovery period for both protocols (all $P_s < 0.05$, $d = 0.85$ to 0.95).

3.4. Changes in psychological conditions for the stroop tests throughout experimental session

Changes in psychological conditions for the Stroop tests throughout LV-HIIE and MCE sessions are shown in Table 3. Arousal analysis revealed a significant main effect for time ($F_{(1.74,33.06)} = 20.77$, $P < 0.001$, $\eta_p^2 = 0.51$) and a significant interaction effect ($F_{(2.81,53.47)} = 3.01$, $P = 0.041$, $\eta_p^2 = 0.14$). Arousal significantly increased immediately after both LV-HIIE and MCE compared with that before each exercise (both $P_s < 0.001$, $d = 2.00$ and 1.90 , respectively), and the increased arousal

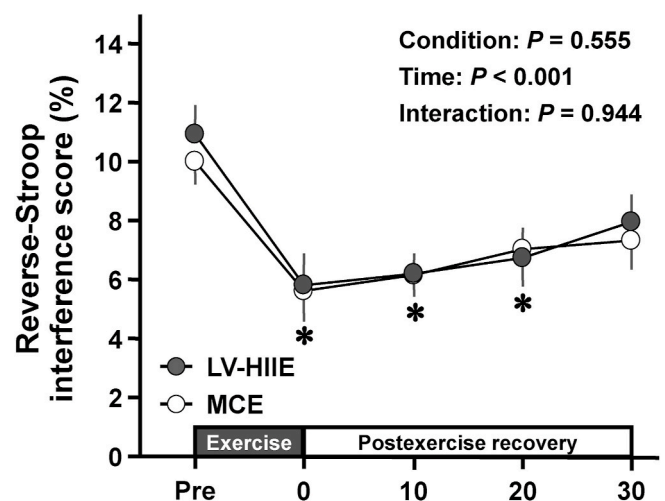


Fig. 2. Changes in IC throughout experimental session in LV-HIIE and MCE. Values are presented as Mean ± SEM. Changes in reverse-Stroop interference scores (i.e., IC) are presented using solid circles for LV-HIIE and open circles for MCE. The reverse-Stroop interference scores analysis revealed significant main effect for time ($P < 0.001$). * Significant difference ($P < 0.05$) from Pre (i.e., before exercise) for both LV-HIIE and MCE.

remained significant until the 20- and 30-min post-exercise recovery periods for LV-HIIE and MCE, respectively (all $P_s < 0.05$, $d = 0.68$ to 0.75). Mental fatigue analysis revealed a significant effect for time ($F_{(2.05,38.88)} = 52.91$, $P < 0.001$, $\eta_p^2 = 0.74$). Mental fatigue significantly increased immediately after both LV-HIIE and MCE compared with that before each exercise (both $P_s < 0.001$, $d = 3.25$ and 3.22 , respectively), and the increased mental fatigue remained significant until the 20-min post-exercise recovery period for both protocols (all $P_s < 0.01$, $d = 1.16$ to 1.25). A trend against significance main effect was observed for time in concentration analysis ($F_{(2.02,28.35)} = 3.00$, $P = 0.061$, $\eta_p^2 = 0.14$).

4. Discussion

In this study, we hypothesized that the degree of post-exercise IC improvements would be greater following LV-HIIE than following MCE. The present findings determined that the degree of IC improvements throughout the 30-min post-exercise period was similar between the two

Table 3

Changes in psychological conditions for the Stroop test throughout LV-HIIE and MCE sections.

	Time points					P values		
	Pre-EX	Post-EX 0	Post-EX 10	Post-EX 20	Post-EX 30	Condition	Time	Interaction
Arousal								
LV-HIIE	2.4 ± 0.2	4.5 ± 0.3 ^a	3.6 ± 0.3 ^{ab}	3.1 ± 0.2 ^{abc}	3.0 ± 0.2 ^{bc}	0.878	<0.001	0.041
MCE	2.6 ± 0.2	4.3 ± 0.2 ^a	3.4 ± 0.2 ^{ab}	3.2 ± 0.2 ^{ab}	3.1 ± 0.2 ^{ab}			
Mental fatigue (mm)								
LV-HIIE	16.7 ± 3.8	72.2 ± 3.9 ^a	49.4 ± 4.4 ^{ab}	38.0 ± 4.4 ^{abc}	33.0 ± 4.7 ^{bc}	0.994	<0.001	0.269
MCE	18.4 ± 3.3	67.0 ± 3.4 ^a	48.3 ± 4.0 ^{ab}	41.0 ± 4.6 ^{abc}	34.5 ± 4.8 ^{bcd}			
Concentration (mm)								
LV-HIIE	69.7 ± 3.6	75.0 ± 3.7	66.8 ± 3.8	63.7 ± 4.3	65.8 ± 3.9	0.509	0.061	0.330
MCE	64.0 ± 3.3	73.4 ± 4.2	69.8 ± 3.4	65.5 ± 4.0	62.3 ± 3.8			
Motivation (mm)								
LV-HIIE	73.0 ± 3.5	76.8 ± 3.7	69.9 ± 3.9	69.6 ± 3.8	73.3 ± 3.5	0.534	0.297	0.083
MCE	68.5 ± 2.6	74.9 ± 4.0	73.2 ± 3.2	71.0 ± 3.6	70.7 ± 3.7			

Values are presented as Mean ± SEM. Bold P values indicate significant main effect of time and significant interaction effect ($P < 0.05$ for all). ^a Significant difference ($P < 0.05$) from Pre-EX. ^b Significant difference ($P < 0.05$) from Post-EX 0. ^c Significant difference ($P < 0.05$) from Post-EX 10. ^d Significant difference ($P < 0.05$) from Post-EX 20.

protocols; thus, our hypothesis was rejected. Nevertheless, our previous study demonstrated that the MCE protocol employed in this study is the most effective in improving IC among five different continuous exercise protocols (Tsukamoto et al., 2017). Therefore, the present findings suggest that LV-HIIE is an effective protocol for stimulating IC improvements, despite lower exercise volume and shorter exercise duration compared to an optimal MCE protocol.

To the best of our knowledge, two previous studies have examined the effect of HIIE on post-exercise IC by using an HIIE protocol with a volume similar to that of the LV-HIIE protocol employed in the present study. Alves et al. (2014) employed an HIIE cycling protocol consisting of ten 1-min bouts at 80% HR reserve and 1 min active recovery at 60% HR reserve and reported that immediate post-exercise IC improvements were greater following the HIIE protocol than following stretching exercise. Kao, Drollette, Ritondale, Khan, and Hillman (2018) employed an HIIE running protocol consisting of eight 1-min bouts at 90% HR max with 1-min active recovery of self-paced walking and determined that the HIIE protocol may be more effective in improving IC than volume-matched MCE (i.e., 20-min bout at 70% HR max). Nevertheless, prior to the present study, the effect of a HIIE protocol (i.e., LV-HIIE) with a comparatively lower volume than MCE on post-exercise IC improvements was unknown. Additionally, we previously determined that although the degree (i.e., immediate effect) of improvement in IC immediately after exercise was similar between HV-HIIE and MCE, the duration (i.e., sustainable effect) of significant improvements in IC throughout the 30-min post-exercise recovery period was longer following HV-HIIE than following MCE (Tsukamoto et al., 2016a), suggesting that the measurements of both immediate and sustainable effects on post-exercise IC improvements may be useful in determining the underlying potential of the effective exercise protocols to improve IC. Therefore, the present study is the first to determine the potential impact of LV-HIIE on post-exercise IC improvements by showing that LV-HIIE and MCE had similar immediate and sustainable effects throughout the 30-min post-exercise recovery period.

An exercise-induced increase in circulating lactate levels plays an important role in improving IC (Hashimoto et al., 2018; Tsukamoto et al., 2016a,b), potentially by increasing cerebral lactate metabolism (Hashimoto et al., 2018; van Hall et al., 2009). In this study, we determined that blood lactate levels throughout the 30-min post-exercise recovery period were higher following LV-HIIE than following MCE. This present finding corroborates the result of previous studies (Parker et al., 2017; Poon et al., 2018). Despite the greater advantage of LV-HIIE in improving post-exercise IC, the degree of post-exercise IC improvements was similar between LV-HIIE and MCE. Our previous study reported that post-exercise IC improvements following MCE with 20-min

duration were sustained for a shorter period of time than that following MCE with 40-min duration, despite higher lactate levels during post-exercise recovery periods (Tsukamoto et al., 2017). These findings suggest that an increase in exercise duration may contribute to determining the degree of post-exercise IC improvements. The acceleration of cerebral lactate metabolism is recognized within 10 min after the start of MCE (Dalsgaard, Ide, Cai, Quistorff, & Secher, 2002; Ide, Horn, & Secher, 1999). Therefore, both duration and amount of lactate metabolized in the brain during exercise may play important roles in improving post-exercise IC. This interpretation may help in understanding the present findings that the degree of post-exercise IC improvements was similar between LV-HIIE and MCE.

Previous studies reported that cognitive functions at rest were better with a higher blood glucose level following breakfast consumption than with a lower blood glucose level following breakfast omission (Komiya et al., 2016; Cooper, Bandelow, & Nevill, 2011). Thus, blood glucose levels may be related to improvement in cognitive function following aerobic exercise. Nevertheless, we previously determined that although the blood glucose level immediately after exercise was higher for HV-HIIE than for MCE, the degree of IC improvement immediately after exercise was similar between the two protocols (Tsukamoto et al., 2016a). In the present study, although the blood glucose levels throughout the 30-min post-exercise recovery period were higher following LV-HIIE than following MCE, the degree of IC improvements throughout the 30-min post-exercise recovery period were similar between the two protocols. Therefore, the difference in the changes in blood glucose levels following HIIE and MCE may not play an important role in determining the degree of their post-exercise IC improvements.

Aerobic exercise-induced changes in psychological parameters, especially arousal and mental fatigue, may play important roles in improving post-exercise IC (Byun et al., 2014; Grego et al., 2005). Byun et al. (2014) reported that an increase in arousal induced by aerobic exercise is related to the degree of immediate post-exercise IC improvements. Furthermore, our previous study demonstrated that increase in arousal after exercise was greater for HV-HIIE than for MCE, which may partially explain greater effect of HV-HIIE than that of MCE on post-exercise IC improvements (Tsukamoto et al., 2016a). In the present study, although a significant interaction effect for arousal was observed, there was no significant difference in arousal throughout experimental session between LV-HIIE and MCE. In addition, although mental fatigue may be associated with the degree of cognitive function improvements after aerobic exercise (Grego et al., 2005), the present study showed no significant difference in mental fatigue throughout experimental session between LV-HIIE and MCE. Therefore, the results of arousal and mental fatigue may, at least partially, explain the similar

improvements in post-exercise IC between the two protocols.

This study has several limitations. Although we determined that a 45-min sitting rest has no effect on changes in IC in a preliminary study, we did not include the control condition in the experimental protocols for the main study. Hence, the main result of this study could not completely exclude some effects (e.g., learning effect) from post-exercise IC improvements induced by LV-HIIE and MCE; thus, this study was focused only on the comparison of the degree of post-exercise IC improvements between the two protocols. In addition, the present study recruited only healthy males, and therefore, it remains unclear whether the present findings can be extrapolated to other populations with different demographics. In particular, decreases in some physical functions, such as exercise intolerance and cardiovascular dysfunction, are potent risks for development and progression of cognitive dysfunction (e.g., mild cognitive impairment or Alzheimer's disease) in older individuals (Defina et al., 2013; Hajjar, Goldstein, Martin, & Quyyumi, 2016). HIIE is recognized as an effective exercise protocol for improving these physical functions in older individuals and/or patients with chronic diseases in addition to healthy adults (Weston, Wisløff, & Coombes, 2014), with a low risk of cardiovascular events during exercise (Rognmo et al., 2012). Furthermore, the circulating level of brain-derived neurotrophic factor (BDNF), a potent marker in the regulation of cognitive function and brain health in humans (Autry & Monteggia, 2012), decreases with aging and in chronic diseases (Erickson et al., 2010; Forlenza et al., 2010; Leckie et al., 2014). Saucedo Martinez et al. (2015) reported that an increase in circulating BDNF secretion induced by HIIE, which has the same protocol as that of LV-HIIE employed in the present study, was greater than that induced by volume-matched MCE. In association with the acute BDNF response, long-term intervention of HIIE may be more effective than that of MCE in increasing circulating BDNF level in older individuals and patients with chronic diseases (O'Callaghan et al., 2019), which may contribute to improvement in cognitive function (Erickson et al., 2011; Leckie et al., 2014). Therefore, effective exercise programs, including LV-HIIE, for improving cognitive function and brain health is more important for older individuals and patients with chronic diseases than for healthy young individuals. Further studies are needed to determine the effect of LV-HIIE on improvements in cognitive functions in older individuals and patients with chronic diseases.

5. Conclusion

This study demonstrated that despite a lower exercise volume and shorter exercise duration, LV-HIIE improves post-exercise IC to a similar degree as MCE. However, we determined that perceived exertion (i.e., RPE) during exercise was higher for LV-HIIE than for MCE. The greater increase in perceived exertion induced by LV-HIIE may result in decreased adherence to this exercise for some individuals (Trost et al., 2002). In a recent study, we reported that listening to music could mitigate the increases in RPE during MCE with 30-min duration, and this effect did not interfere with post-exercise IC improvements (Tanaka et al., 2018). To enhance adherence to LV-HIIE in the clinical setting, further studies are needed to develop effective strategies that would minimize perceived responses during this exercise.

CRedit authorship contribution statement

Takeshi Sugimoto: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Writing - original draft. **Tadashi Suga:** Conceptualization, Data curation, Formal analysis, Investigation, Writing - original draft. **Hayato Tsukamoto:** Data curation, Funding acquisition, Investigation, Methodology, Writing - review & editing. **Thomas A. Calverley:** Investigation, Supervision, Writing - review & editing. **Daichi Tanaka:** Data curation, Investigation. **Saki Takenaka:** Data curation, Investigation. **Keigo Tomoo:** Data curation, Investigation. **Kento Dora:** Data curation, Investigation. **Damian M.**

Bailey: Investigation, Supervision, Writing - review & editing. **Tadao Isaka:** Funding acquisition, Investigation, Methodology, Project administration, Writing - review & editing. **Takeshi Hashimoto:** Funding acquisition, Investigation, Methodology, Project administration, Writing - review & editing.

Declaration of competing interest

The authors declare no conflict of interest.

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